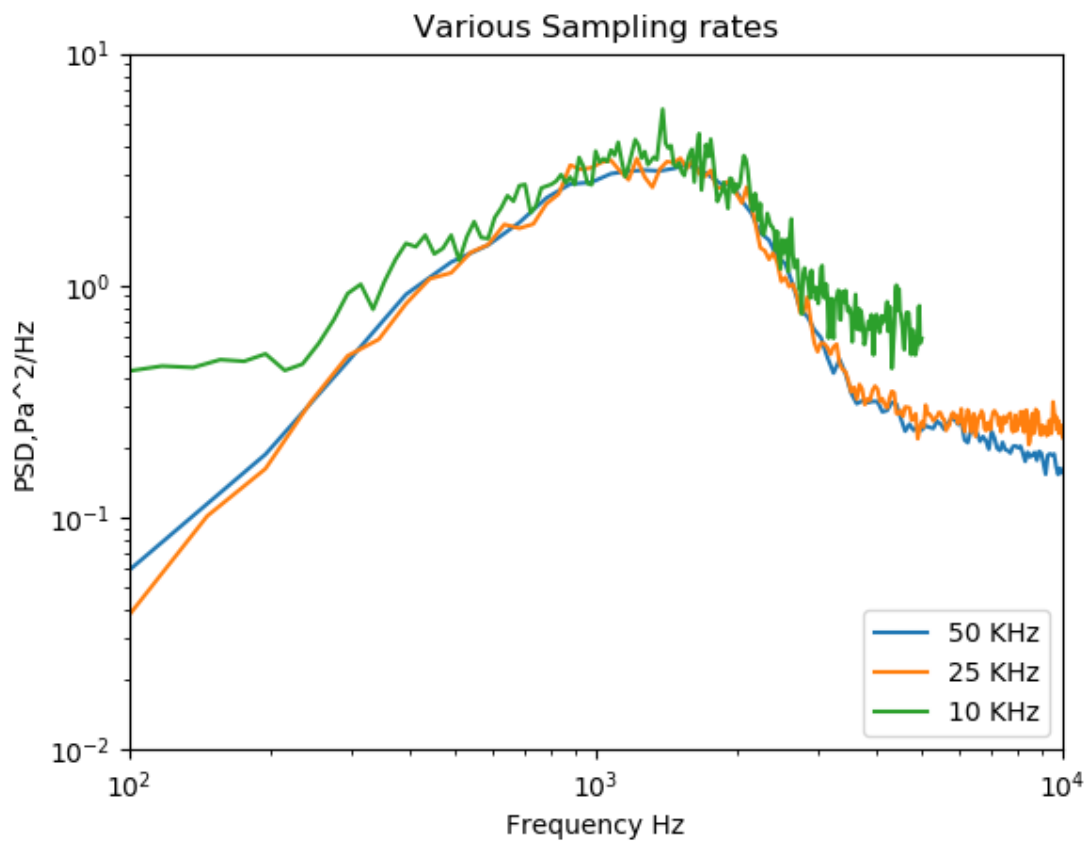
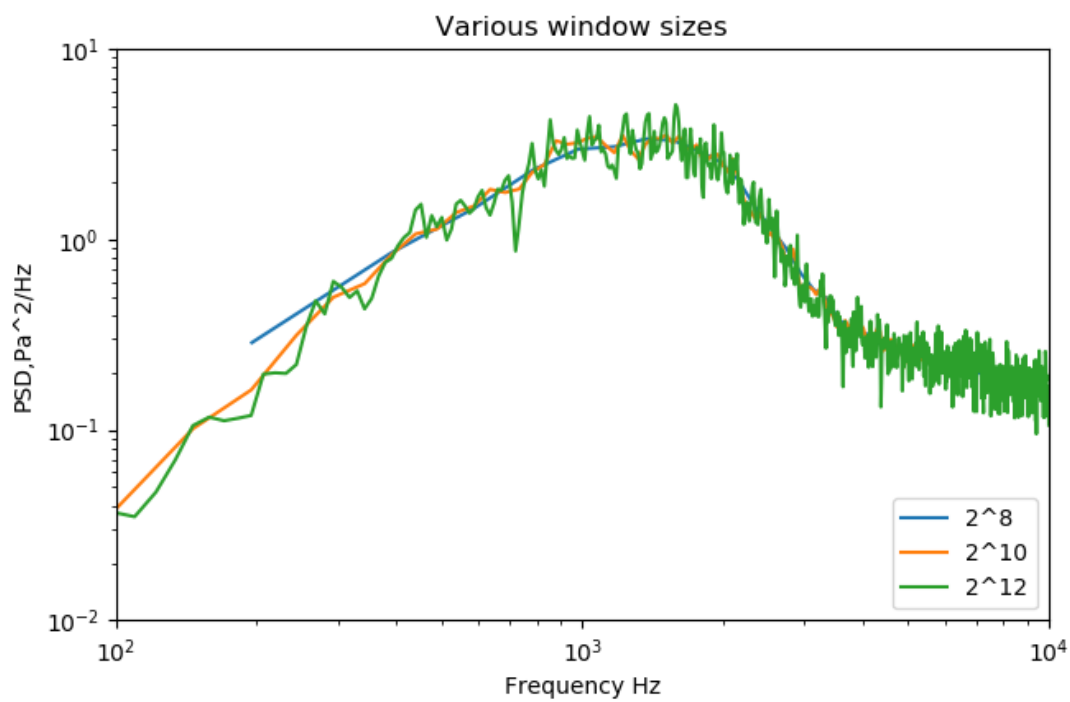




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import numpy as np
import math as mh
import cmath
import matplotlib.pyplot as plt
prsr=np.loadtxt("pressure.txt", delimiter=" ")
def calPSD(prsr,N,fs,w,olp):
    k=range(N)
    L=len(prsr)
    sigma=0
    wk=[]
    if w=="hann":
        for i in range(N):
            tem=0.5 * (1 - mh.cos((2*mh.pi*k[i])/(N-1)))
            wk.append(tem)
            sigma+=(tem)**2
        sigma=(sigma/N)**-0.5

    if w=="hamming":
        for i in range(N):
            tem= 0.54 - 0.46*(mh.cos((2*mh.pi*k[i])/(N-1)))
            wk.append(tem)
            sigma+=(tem)**2
        sigma=(sigma/N)**-0.5

    if w=="rectwin":
        for i in range(N):
            tem=1
            wk.append(tem)
            sigma+=(tem)**2
        sigma=(sigma/N)**-0.5
    chi=N/fs
    olp_int=round(N*(100-olp)/100)
    olp_tem=0
    olp_index=[]
    while olp_tem<L-N+1:
        olp_index.append(olp_tem)
        olp_tem+=olp_int
    print(olp_index)
    F=np.linspace(0,fs,N)
    PSDg=[]
    def Expect(n):
        El=0

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    lth=len(olp_index)
    for i in olp_index:
        flk=prsr[i:(i+N)]
        flk=flk-np.average(flk)
        comp=0
        for j in range(N):
            modu=cmath.rect(wk[j]*flk[j],-2*mh.pi*n*j/N)
            comp+=modu
        comp=comp/N
        Ek=cmath.polar(comp)
        Ek=(Ek[0])**2
        El+=Ek
    El=El/lth
    return El
for i in range(round(len(F)/2)):
    n=i
    tem=(2-1//((i+1)))*chi*(sigma**2)*Expect(n)
    PSDg.append(tem)

return (PSDg,F)

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## Various Window Size
figure_1=plt.figure(1)
(PSDg,F)=calPSD(prsr,256,50000,"hann",10)
F=F[1:round(len(F)/2)]
PSDg=PSDg[1:]
plt.plot(F,PSDg,label='2^8')
(PSDg,F)=calPSD(prsr,1024,50000,"hann",10)
F=F[1:round(len(F)/2)]
PSDg=PSDg[1:]
plt.plot(F,PSDg,label='2^10')
(PSDg,F)=calPSD(prsr,4096,50000,"hann",10)
F=F[1:round(len(F)/2)]
PSDg=PSDg[1:]
plt.plot(F,PSDg,label='2^12')
plt.title("Various window sizes")
plt.ylabel("PSD,Pa^2/Hz")
plt.xlabel("Frequency Hz")
plt.legend(loc='lower right')
plt.xscale("log")
plt.yscale("log")
plt.xlim(100,10000)

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plt.ylim(0.01,10)
figure_1.savefig('window_size.png')
## various sampling rates
figure_2=plt.figure(2)
(PSDg,F)=calPSD(prsr,512,50000,"hann",10)
F=F[1:round(len(F)/2)]
PSDg=PSDg[1:]
plt.plot(F,PSDg,label='50 KHz')
prsr_25=[]
for i in range(int(len(prsr)/2)):
    prsr_25.append(prsr[2*i])
(PSDg,F)=calPSD(prsr_25,512,25000,"hann",10)
del prsr_25
F=F[1:round(len(F)/2)]
PSDg=PSDg[1:]
plt.plot(F,PSDg,label='25 KHz')
prsr_10=[]
for i in range(int(len(prsr)/5)):
    prsr_10.append(prsr[5*i])
(PSDg,F)=calPSD(prsr_10,512,10000,"hann",10)
del prsr
del prsr_10
F=F[1:round(len(F)/2)]
PSDg=PSDg[1:]
plt.plot(F,PSDg,label='10 KHz')
plt.xscale("log")
plt.yscale("log")
plt.xlim(100,10000)
plt.ylim(0.01,10)
plt.title("Various Sampling rates")
plt.ylabel("PSD,Pa^2/Hz")
plt.xlabel("Frequency Hz")
plt.legend(loc='lower right')
figure_2.savefig('sample_rates.png')

```